

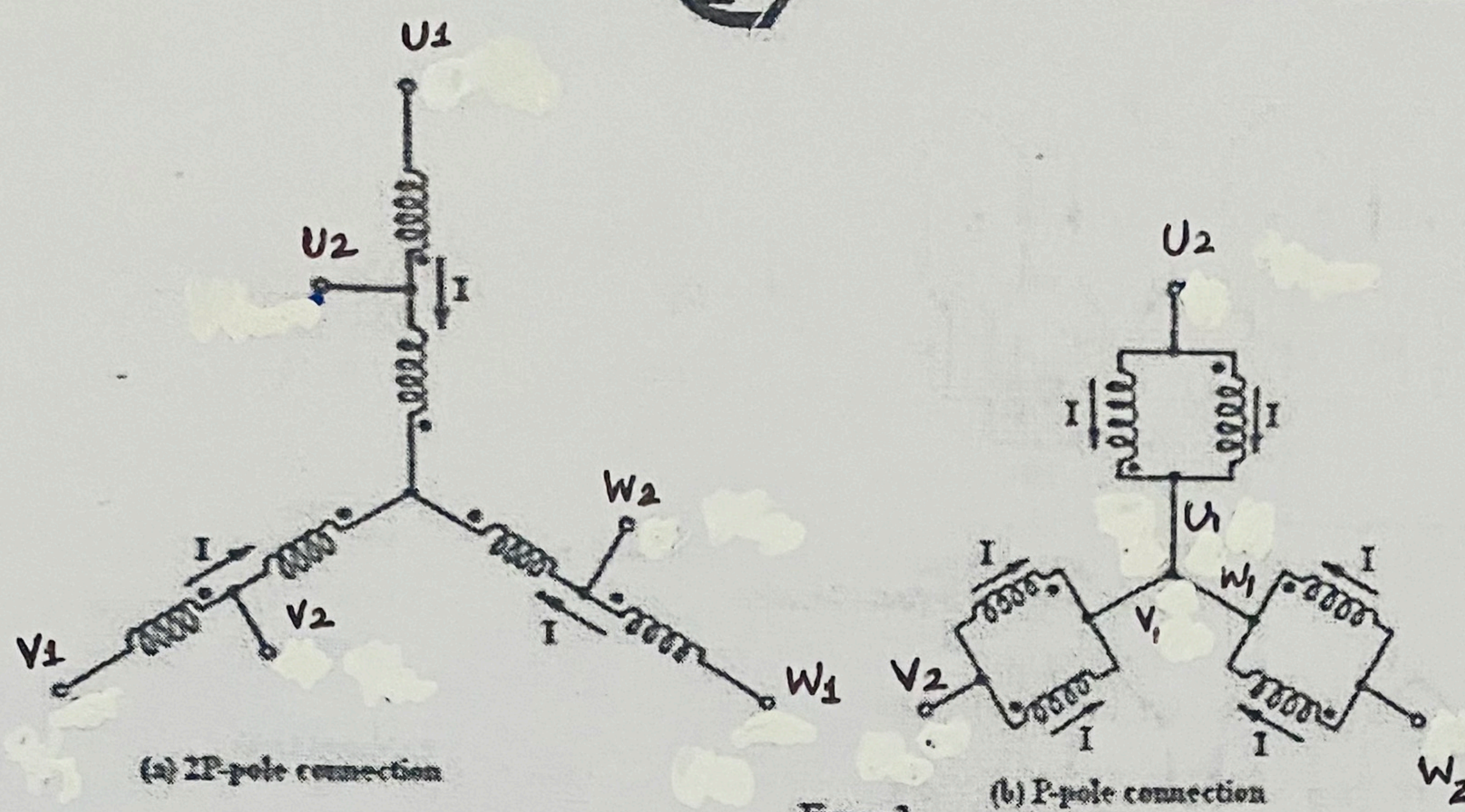


Figure 2

It is assumed that the power factor remains unchanged and the motor losses are negligible. With the changeover of stator winding from series-star to parallel-star, the power drawn from the supply is doubled. Simultaneously, the speed is also doubled. So, the motor torque remains constant.

b) Constant power type

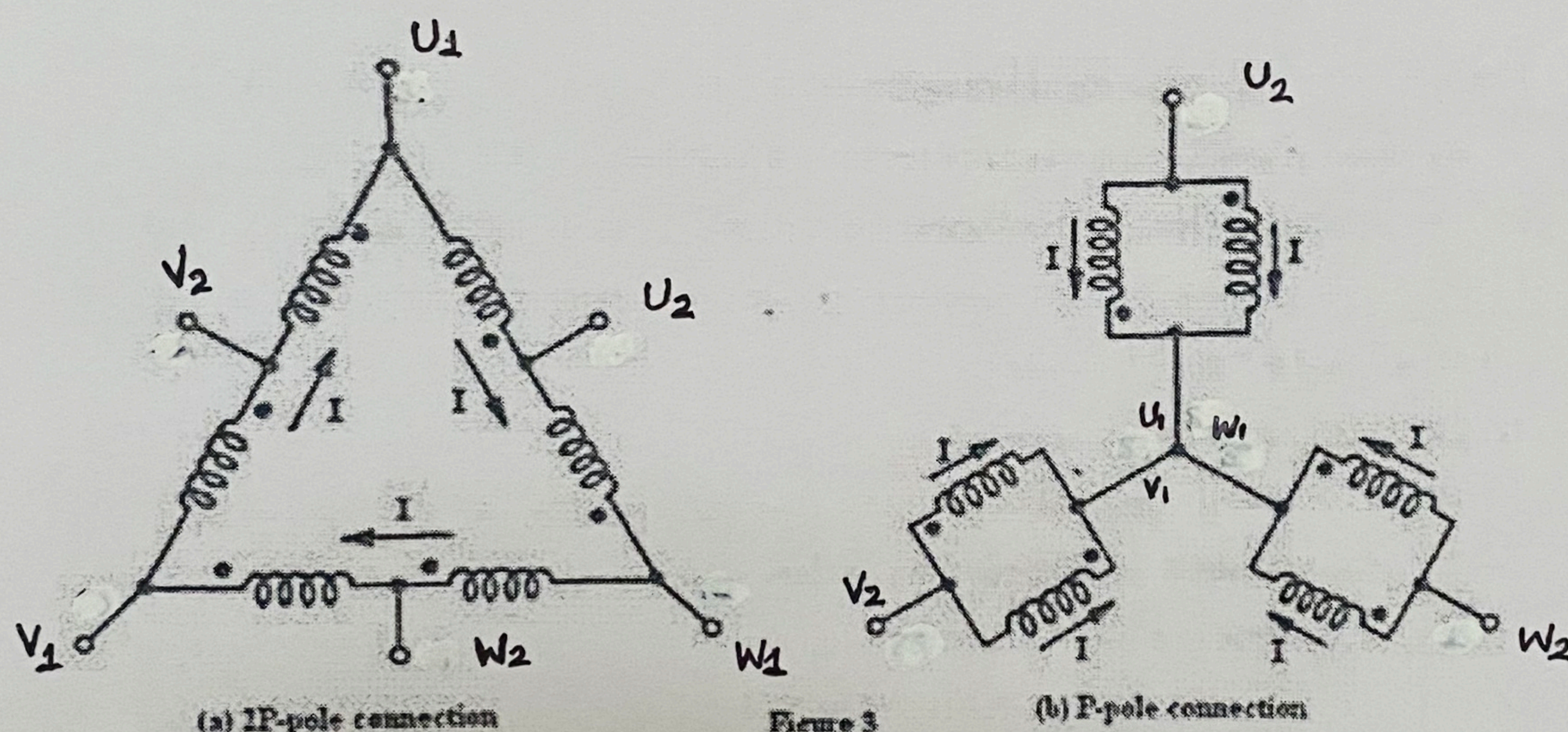


Figure 3

If constant power output is to be obtained from the motor at both the speeds, i.e. the torque is inversely proportional to the speed, the connection of the winding may be done as shown in figure 3. For low speed operation, the voltage is applied to terminals $U_1-V_1-W_1$ and terminals $U_2-V_2-W_2$ are left open, while for high speed operation, the voltage is applied to terminals $U_2-V_1-W_1$ and $V_1-U_1-W_1$ are shorted.

For series – delta connection, $P_{\Delta} = 3VI \cos \phi_{\Delta}$

For parallel – star connection, $P_Y = 2\sqrt{3}VI \cos \phi_{YY} = 3.46VI \cos \phi_{YY}$

After changeover from series-delta to parallel-star, the power increases slightly (about 15%), if power factor is assumed to remain constant. The constant power connection is the most expensive, because in this case the motor size becomes the largest.